ETSI 300-328
Technical Report

RF measurement of STM32WBA

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Script used: BLE_ETSI_300_328.py
Version: v1.5.0

Please Read Carefully:
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The script was executed with result: **All tests passed**

All tests were carried out according to requirements specified in ETSI 300 328 v2.2.2.

Note: In spurious domain, measurements were conducted up to the 12.75 GHz, but the instruments used are defined only up to 8.0 GHz. For that reason, such frequency bands are colored gray and their purpose is only informational.

Author's comment:

Device Output Power: 0.9 dBm (PA value 26)
Device PHY: 1 Ms/s

Hardware used for measurements

Instrument used	Model	Last Calibrated	Max Frequency
Spectral Analyser	R&S FSVA3030	2023-05-23	30 GHz
Cable	-	(-)	8 GHz
Connector	-	(-)	8 GHz

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1) RF Output Power

The limit for RF output power is defined for the mean equivalent isotropic radiated power (e.i.r.p.) of the equipment during a transmission burst.
Limit itself is maximum +20 dBm peak output power. The measurement is carried out in conducted mode, therefore the antenna's gain need to be taken into account for final evaluation.

Table 1. RF output power measured values

Channel frequency (MHz)	Limit (dBm)	Cable Loss (dB)	Power (dBm)	Test
2402	20	0.6	1.0	**Passed**
2442	20	0.6	0.82	**Passed**
2480	20	0.6	0.44	**Passed**

Fig.1.1 RF output power on frequency 2402 MHz

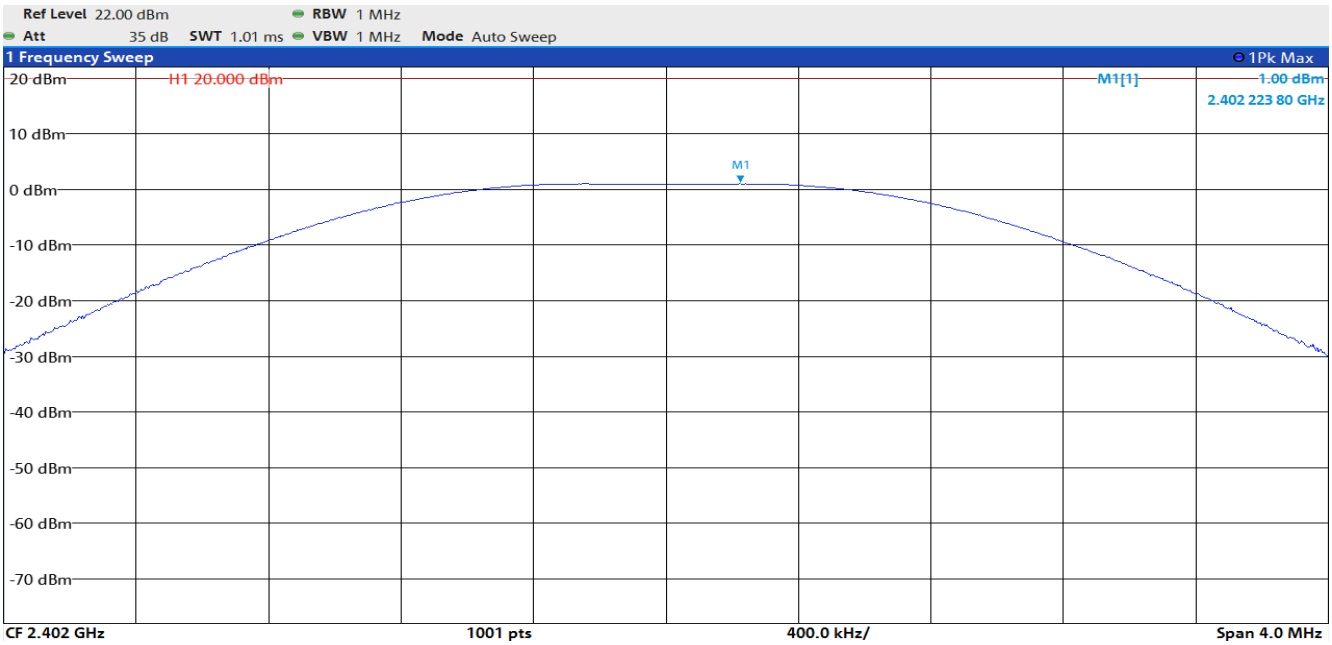


Fig.1.2 RF output power on frequency 2442 MHz

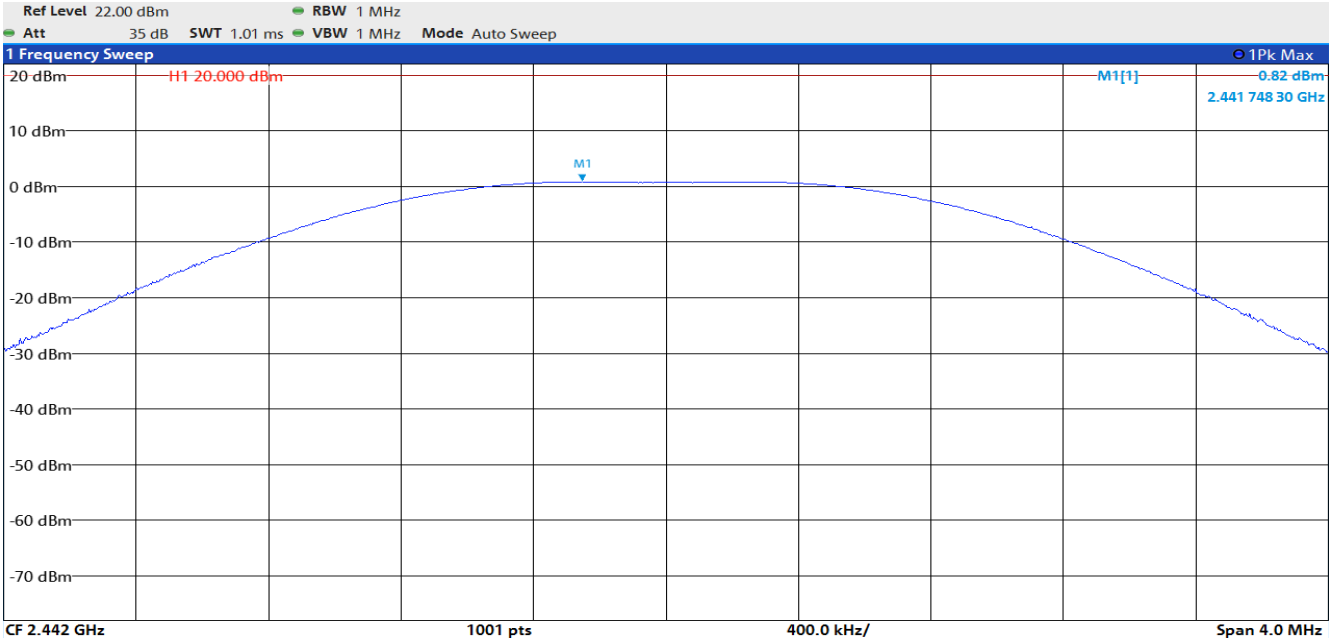
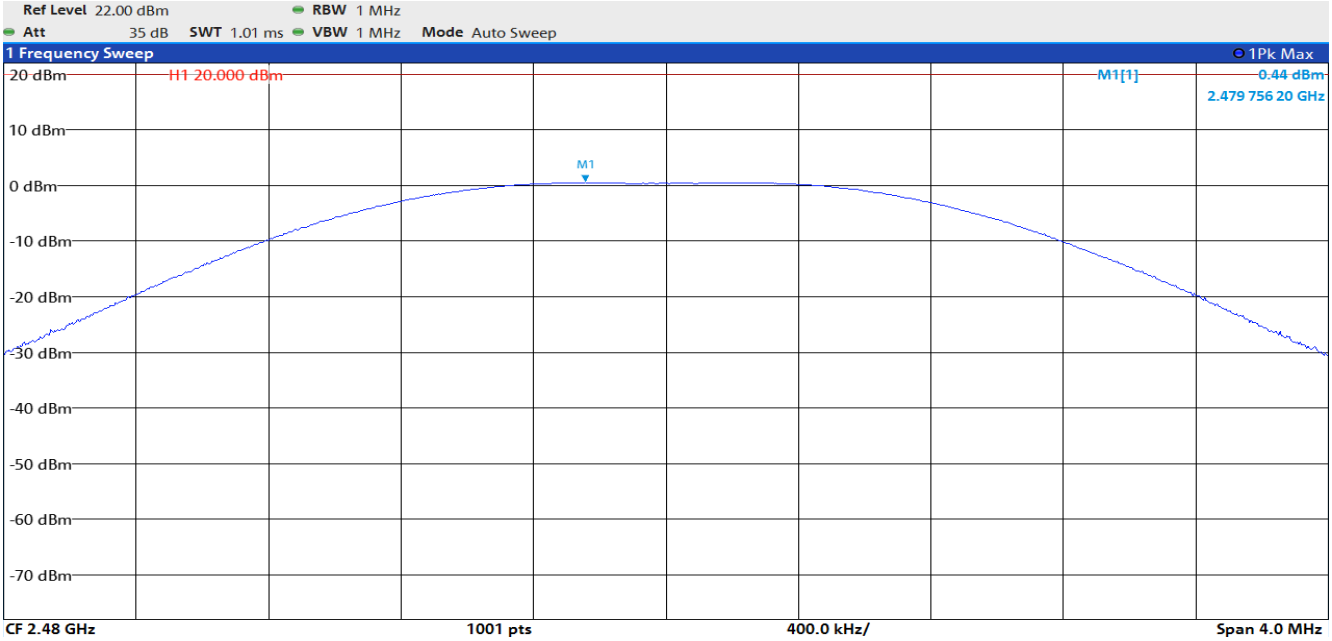


Fig.1.3 RF output power on frequency 2480 MHz



2) Power Spectral density

The Power Spectral Density (PSD) represents power of the signal in frequency segments. In this case, PSD in a 1 MHz bandwidth is measured. The maximum Power Spectral Density per MHz is +10 dBm to comply with the regulation.

Table 2. PSD measured values

Channel frequency (MHz)	Limit (dBm/MHz)	Cable loss (dB)	PSD (dBm/MHz)	Test
2402	10	0.6	1.0	**Passed**
2442	10	0.6	0.8	**Passed**
2480	10	0.6	0.42	**Passed**

Fig.2.1 PSD on frequency 2402 MHz

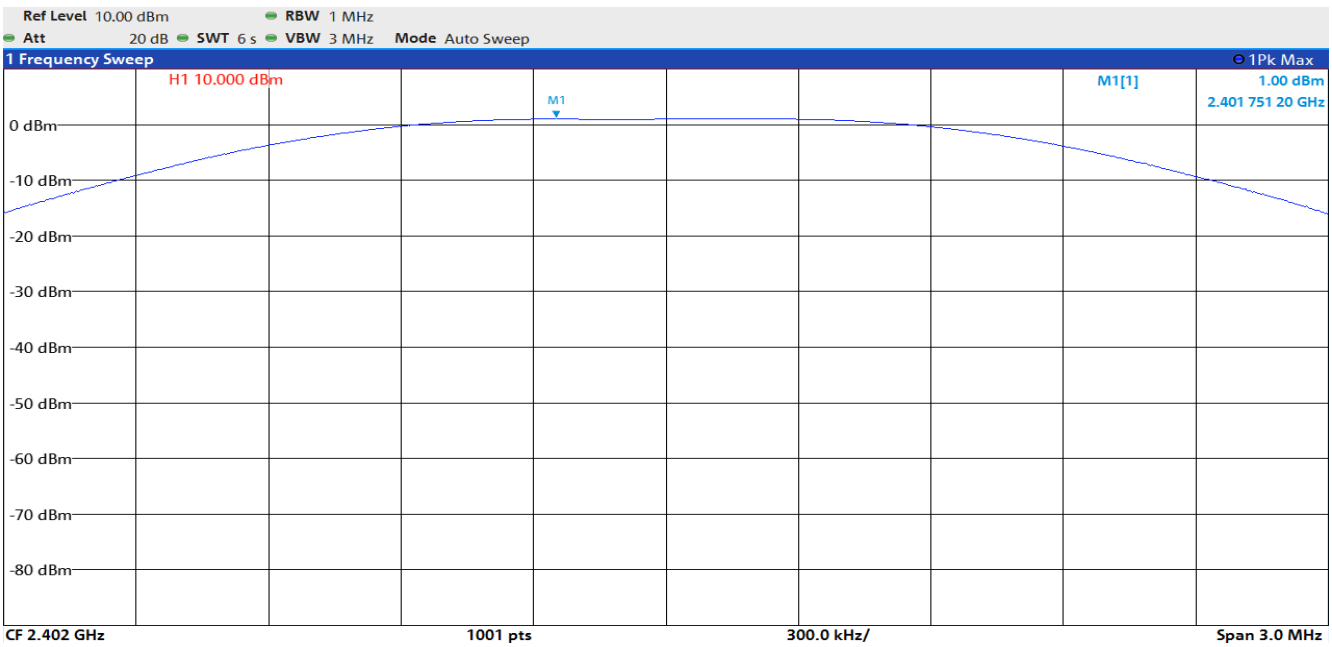
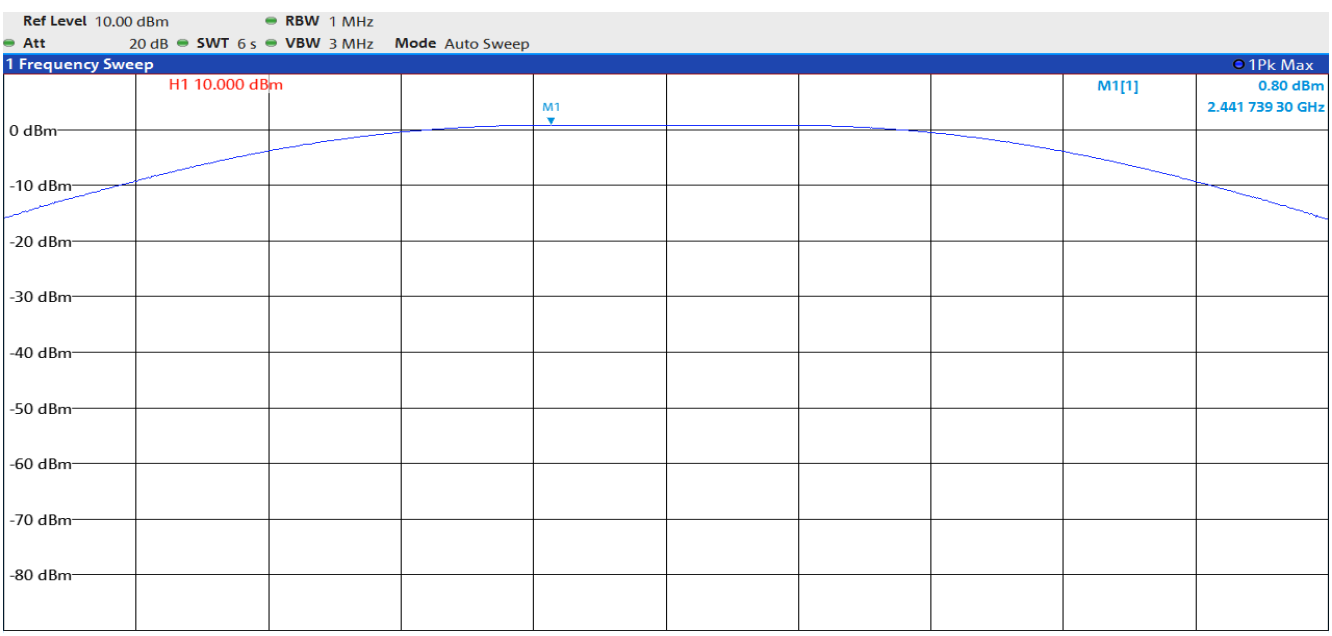


Fig.2.2 PSD on frequency 2442 MHz



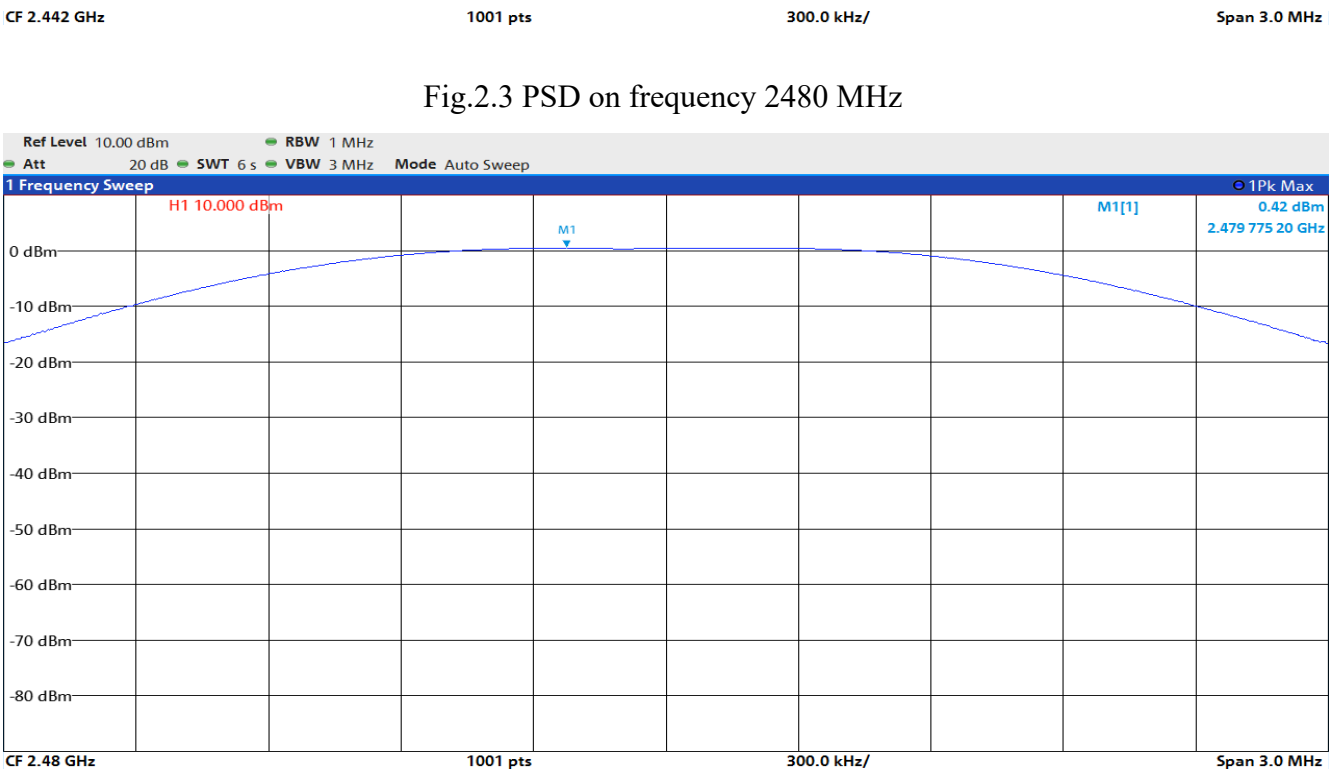


Fig.2.3 PSD on frequency 2480 MHz

3) Occupied Channel Bandwidth

The Occupied Channel Bandwidth is the bandwidth that contains 99 % of the power of the signal.
Limit is defined as 99 % of power that must completely fall within allocated band from 2400 to 2483.5 MHz.
Using OCC function are measured outer frequencies, those are compared with the limits.

Table 3. 99 % occupied bandwidth measured values

Channel frequency (MHz)	OBW (MHz)	T1 (MHz)	T1 (dBm)	T2 (MHz)	T2 (dBm)	Test
2402	1.09	2401.47	-27.1	2402.55	-27.0	**Passed**
2480	1.07	2479.47	-27.5	2480.55	-27.0	**Passed**

Fig.3.1 Occupied 99 % Channel BW on 2402 MHz

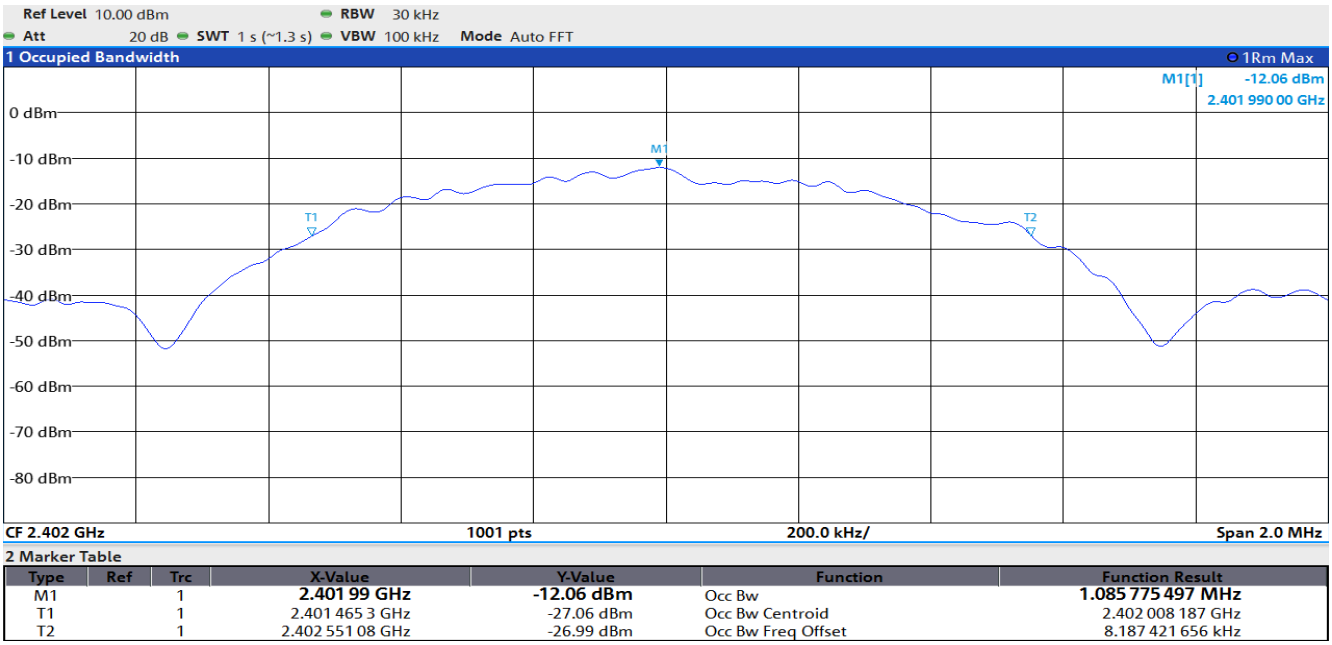
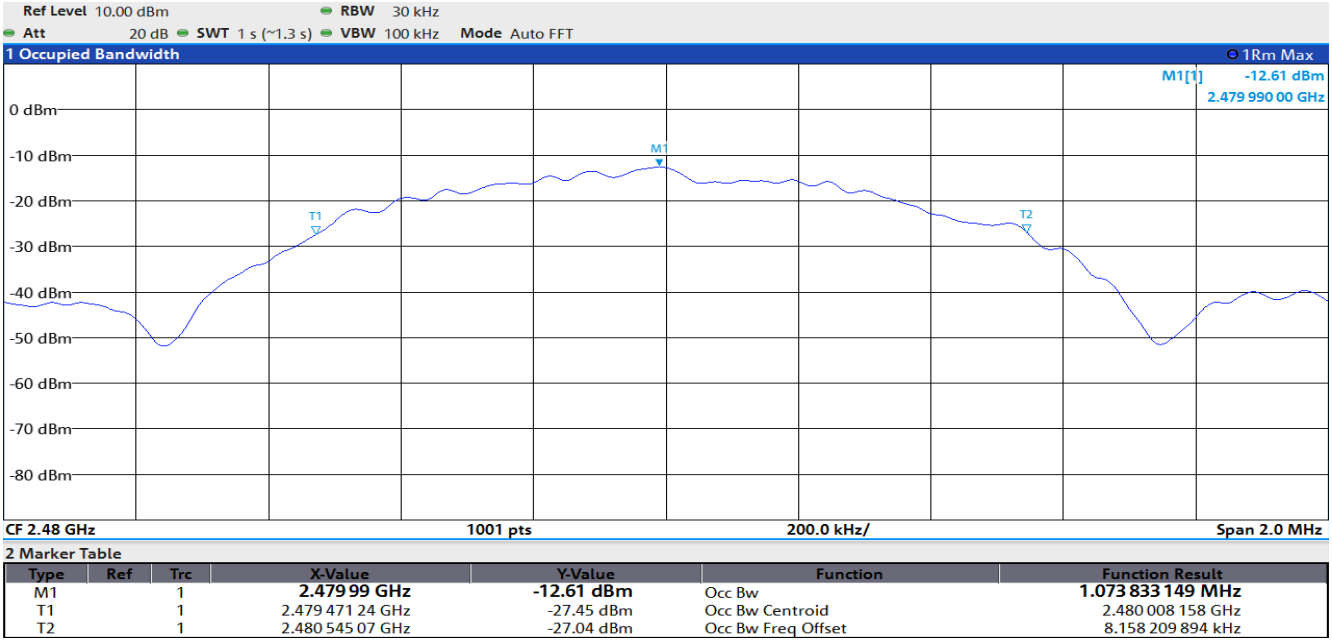


Fig.3.2 Occupied 99 % Channel BW on 2480 MHz



4) Out-of-Band Unwanted Emissions

In ETSI 300 328 v2.2.2 specification, transmitter unwanted emissions in the out-of-band domain are emissions when the equipment is in Transmit mode, on frequencies immediately outside the allocated band, but excluding unwanted emissions in the spurious domain.
The transmitter unwanted emissions in the out-of-band domain shall not exceed -10 dBm/MHz in 2 MHz span around the allocated band and -20 dBm/MHz in 2 to 4 MHz span around the allocated band.

Table 4. Measured emissions in out-of-band domain

Frequency (MHz)	Limit (dBm/MHz)	Cable loss (dB)	Amplitude (dBm/MHz)	Test
Tx on frequency 2402				
2399.5	-10	0.6	-42.26	**Passed**
2399.41	-10	0.6	-42.56	**Passed**
2398.41	-20	0.6	-47.36	**Passed**
2398.32	-20	0.6	-46.37	**Passed**
Tx on frequency 2480				
2484.0	-10	0.6	-49.37	**Passed**
2484.07	-10	0.6	-49.59	**Passed**
2485.07	-20	0.6	-49.63	**Passed**
2485.14	-20	0.6	-50.36	**Passed**

Fig.4.1 Out of Band emission from 2399.0 to 2400.0 MHz

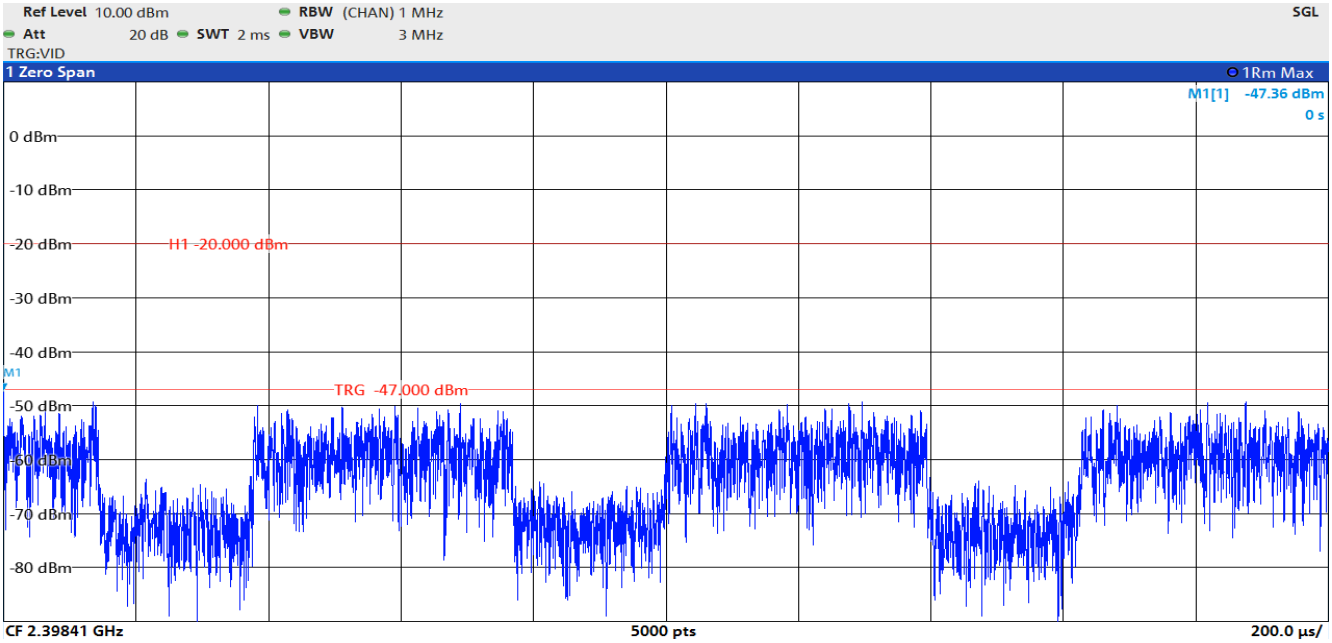


Fig.4.2 Out of Band emission from 2398.91 to 2399.91 MHz

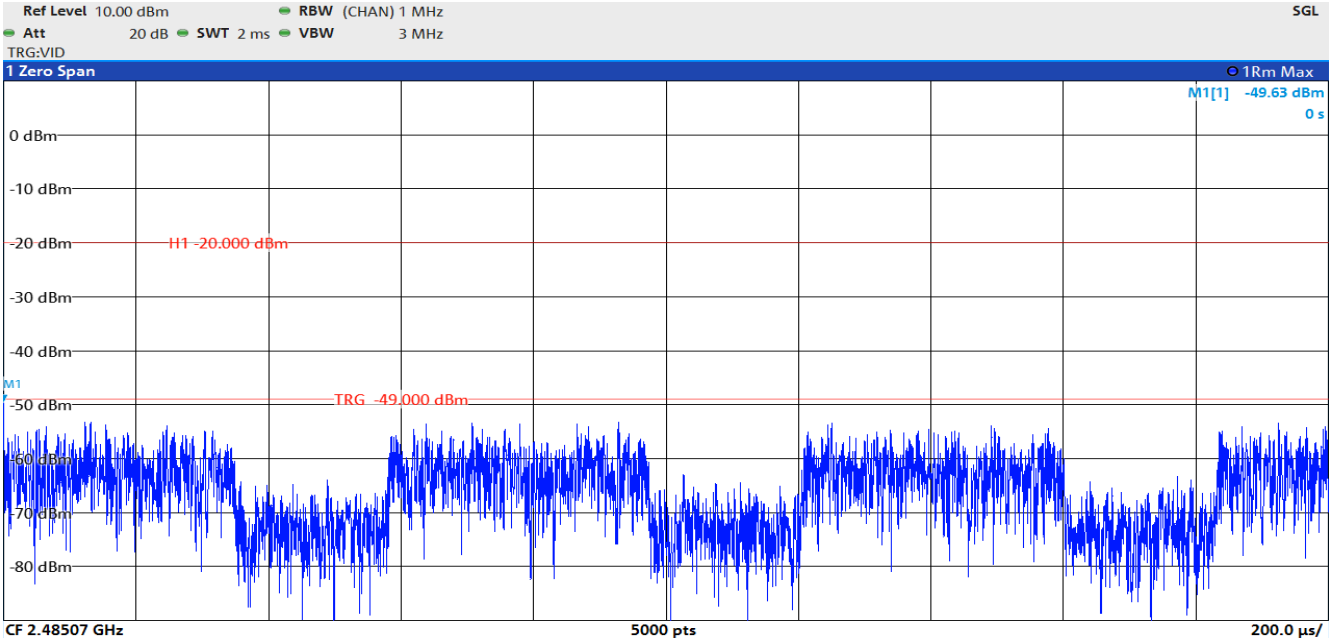


Fig.4.3 Out of Band emission from 2397.91 to 2398.91 MHz

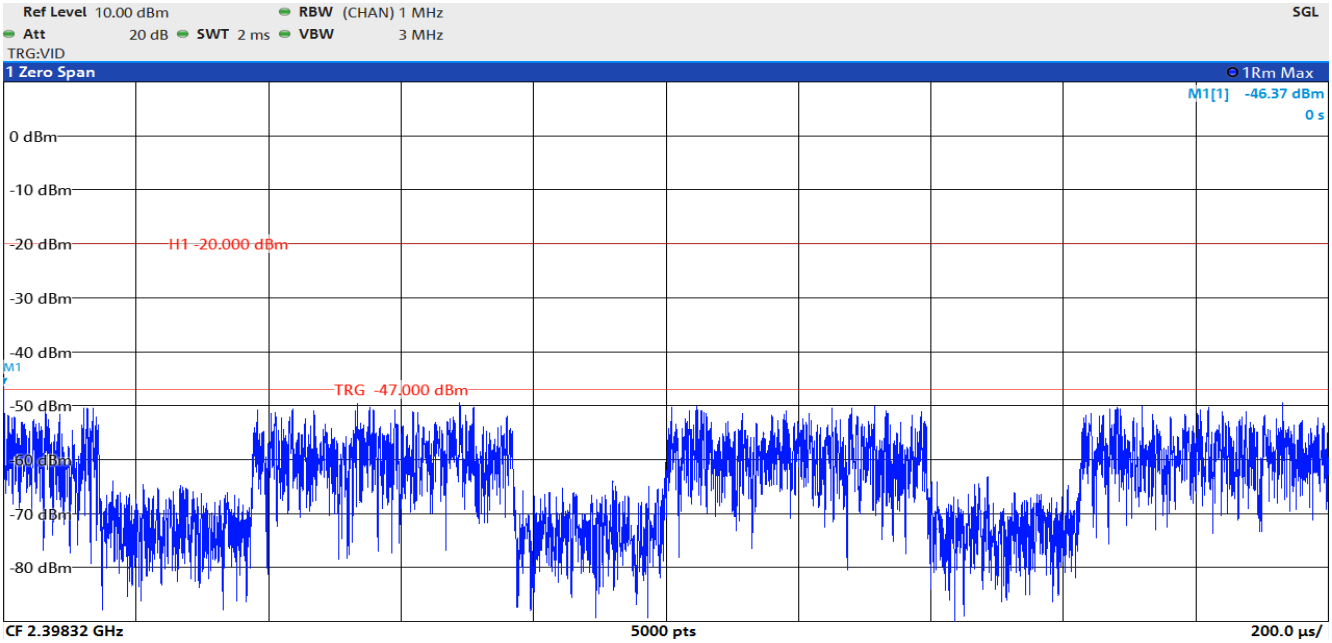
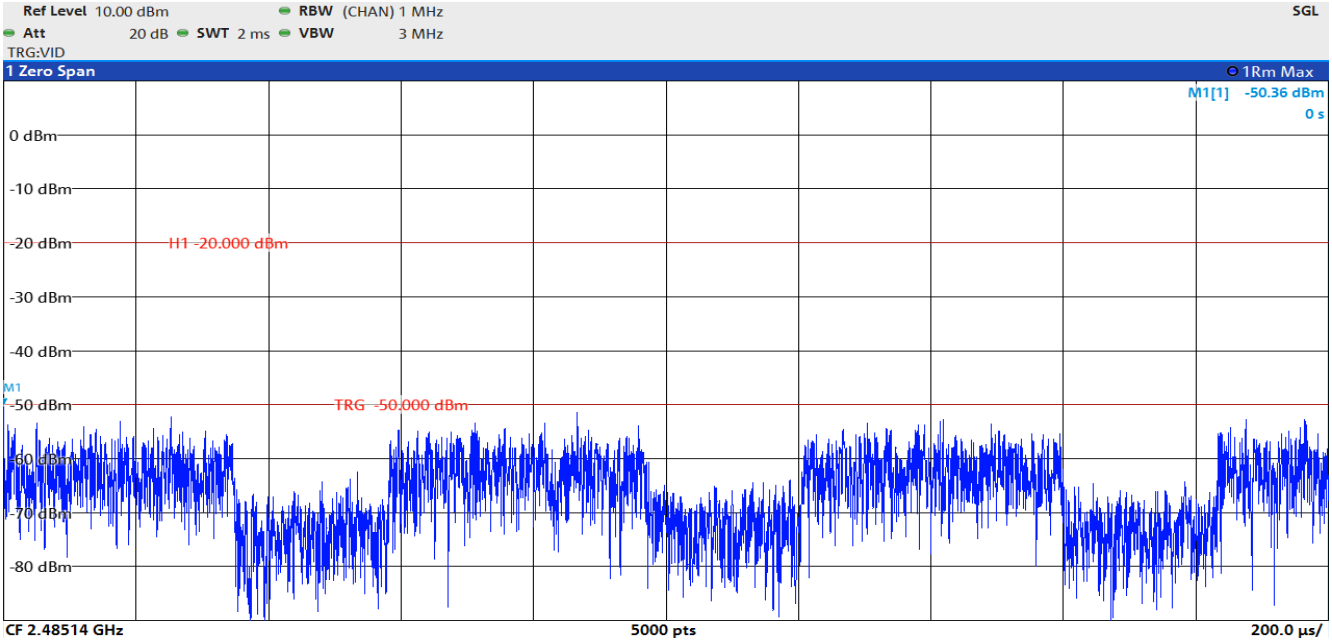


Fig.4.4 Out of Band emission from 2397.82 to 2398.82 MHz



5) Transmitter Spurious Emissions

Transmitter spurious emissions are defined as any radiation outside of an allocated band and outside the out-of-band domain in Tx mode. The measurements are carried out with 100 kHz RBW for sub-GHz frequencies and 1 MHz for frequencies higher than 1 GHz. The limit is set to -54 dBm for restricted bands and -36 in non-restricted bands in sub-GHz frequencies. The limit above 1 GHz is set to -30 dBm.

Table 5. Measured spurious emissions

Frequency Start (MHz)	Frequency Stop (MHz)	Limit (dBm)	RBW (kHz)	Cable Loss (dB)	Measured frequency (MHz)	Measured power (dBm)	Test
Operating frequency: **2402 MHz** and channel **0**							
30	47	-36	100	0.6	42.1	-69.3	**Passed**
47	74	-54	100	0.6	55.8	-70.4	**Passed**
74	87.5	-36	100	0.6	78.3	-70.9	**Passed**
87.5	118	-54	100	0.6	91.8	-69.8	**Passed**
118	174	-36	100	0.6	119.1	-71.0	**Passed**
174	230	-54	100	0.6	177.5	-70.7	**Passed**
230	470	-36	100	0.6	347.8	-71.1	**Passed**
470	694	-54	100	0.6	632.8	-71.1	**Passed**
694	1000	-36	100	0.6	918.2	-70.5	**Passed**
1000	2396	-30	1000	0.6	2395.3	-52.3	**Passed**
2487.5	8000.0	-30	1000	0.6	7468.6	-54.2	**Passed**
8000.0	12750.0	-30	1000	0.6	12159.2	-59.3	***Passed***
Operating frequency: **2480 MHz** and channel **39**							
30	47	-36	100	0.6	41.3	-57.8	**Passed**
47	74	-54	100	0.6	52.4	-61.0	**Passed**
74	87.5	-36	100	0.6	75.6	-59.8	**Passed**
87.5	118	-54	100	0.6	92.3	-60.7	**Passed**
118	174	-36	100	0.6	137.4	-60.2	**Passed**
174	230	-54	100	0.6	203.5	-61.4	**Passed**
230	470	-36	100	0.6	233.5	-60.9	**Passed**
470	694	-54	100	0.6	662.8	-60.0	**Passed**
694	1000	-36	100	0.6	744.9	-60.3	**Passed**
1000	2396	-30	1000	0.6	2364.6	-58.5	**Passed**
2487.5	8000.0	-30	1000	0.6	7490.6	-54.1	**Passed**
8000.0	12750.0	-30	1000	0.6	8348.8	-59.2	***Passed***

Fig.5.1 Tx Spurious emission on 2402 MHz from 150 kHz to 1 GHz

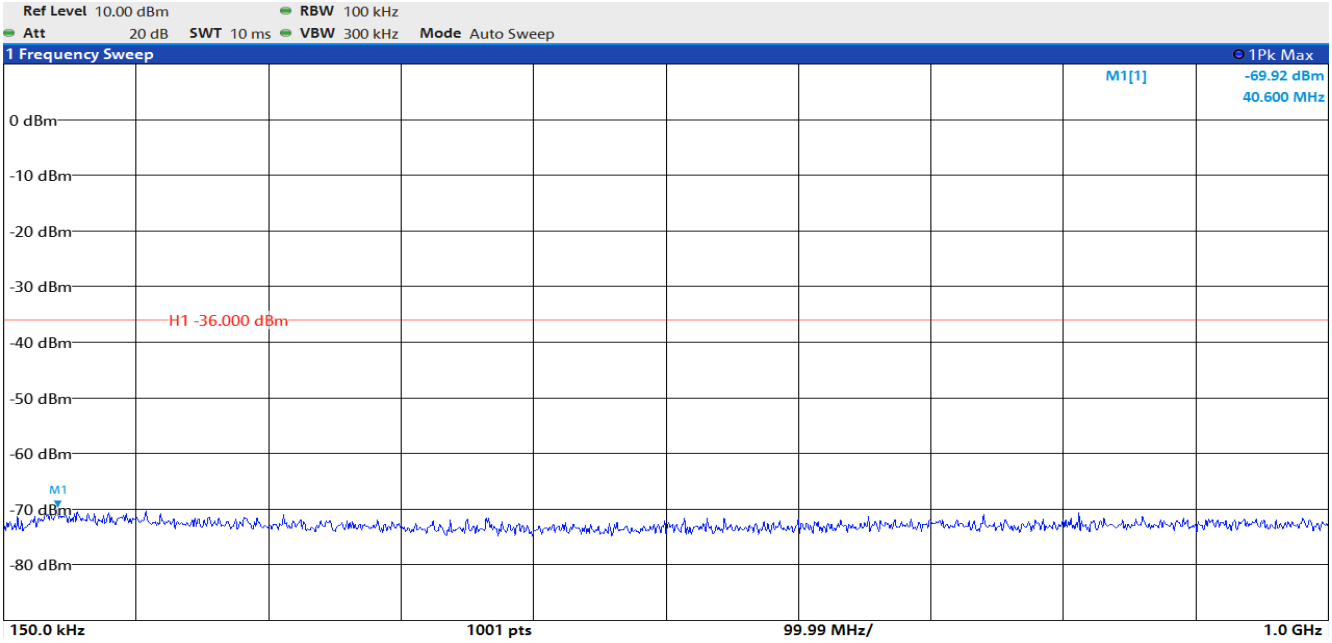


Fig.5.2 Tx Spurious emission on 2402 MHz from 1 to 12.75 GHz

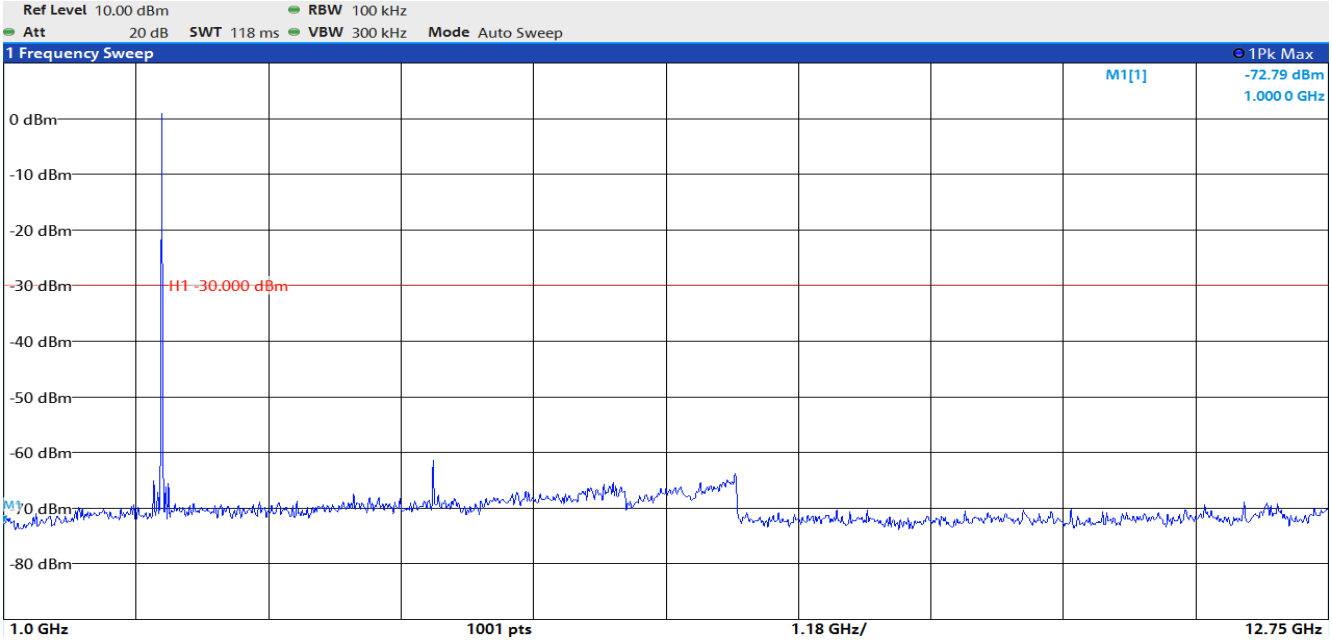


Fig.5.3 Tx Spurious emission on 2440 MHz from 150 kHz to 1 GHz

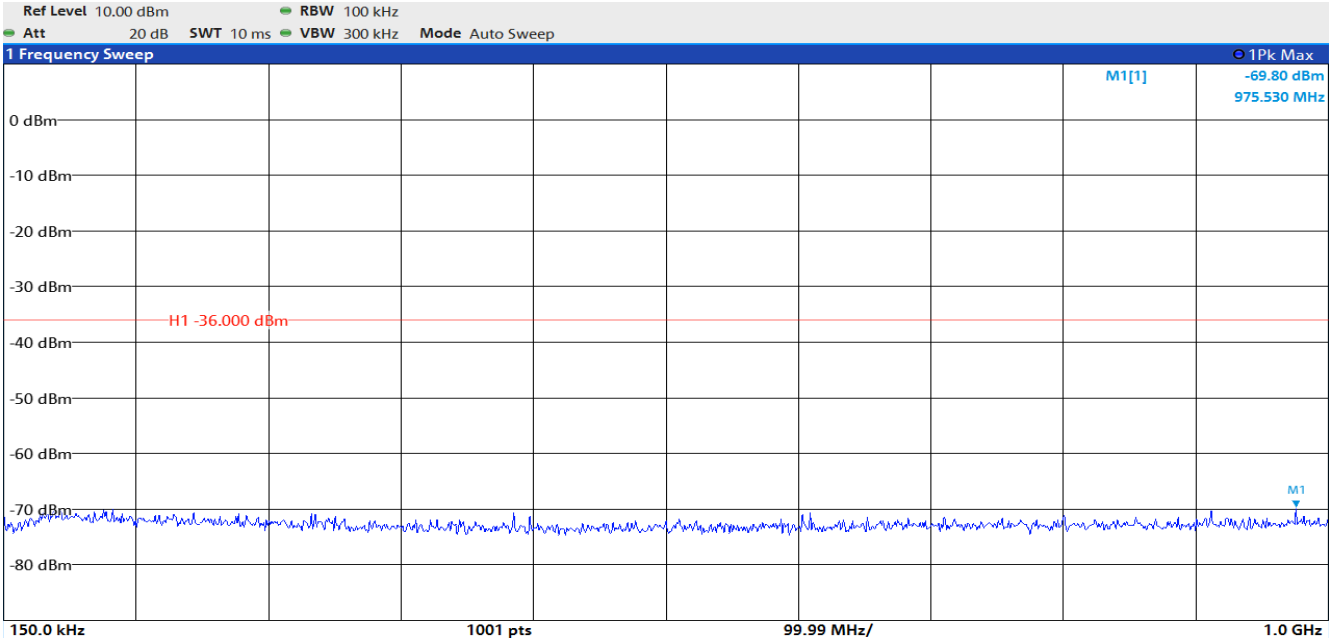
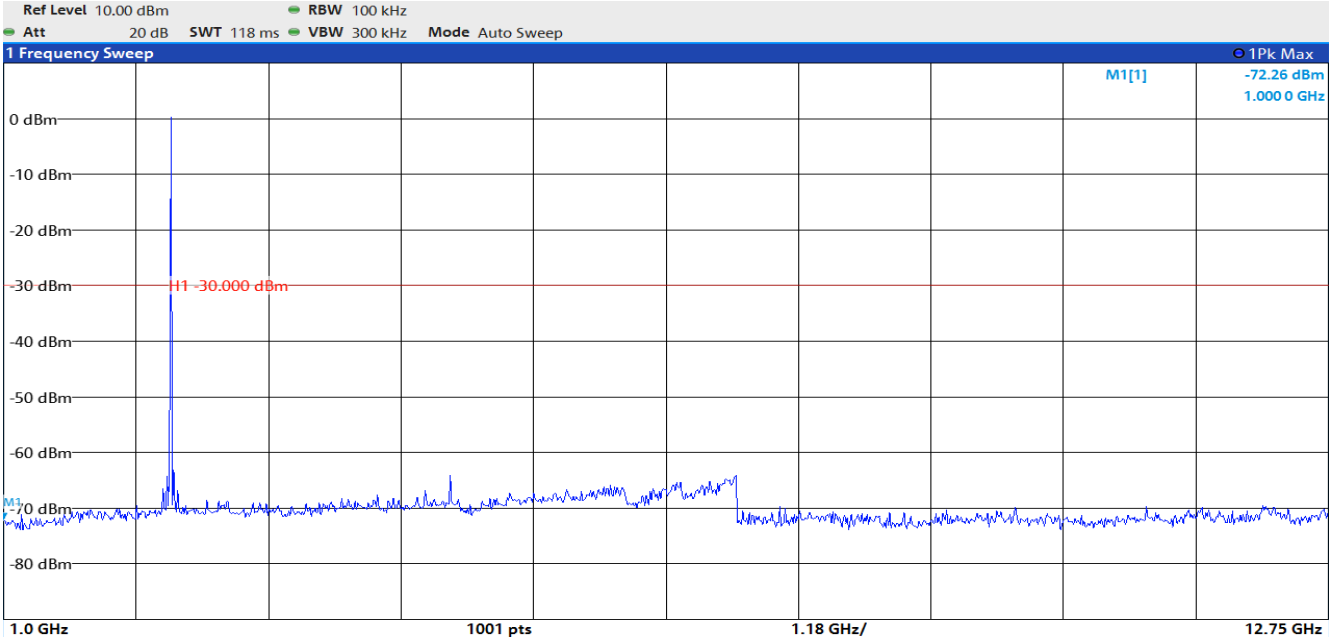


Fig.5.4 Tx Spurious emission on 2440 MHz from 1 to 12.75 GHz



6) Receiver Spurious Emissions

Receiver spurious emissions are defined as any radiation outside of an allocated band and outside the out-of-band domain in Rx mode. The measurements are carried out with 100 kHz RBW for sub-GHz frequencies and 1 MHz RBW for frequencies higher than 1 GHz. The limit is set to -57 dBm for bands in sub-GHz frequencies and -47 dBm above 1 GHz .

Table 6. Measured emissions on the edge of allocated band

Frequency Start (MHz)	Frequency Stop (MHz)	Limit (dBm)	RBW (kHz)	Cable Loss (dB)	Measured frequency (MHz)	Measured power (dBm)	Test
Operating frequency: **2402 MHz, ch 0**							
30	1000	-57	100	0.6	32.4	-80.7	**Passed**
1000	2396	-47	1000	0.6	2250.3	-78.6	**Passed**
2487.5	8000	-47	1000	0.6	6427.8	-74.7	**Passed**
8000	12750.0	-47	1000	0.6	12704.9	-79.1	***Passed***
Operating frequency: **2480 MHz, ch 39**							
30	1000	-57	100	0.6	39.2	-80.0	**Passed**
1000	2396	-47	1000	0.6	1667.3	-78.8	**Passed**
2487.5	8000	-47	1000	0.6	7430.0	-74.4	**Passed**
8000	12750.0	-47	1000	0.6	12230.4	-79.3	***Passed***

Fig.6.1 Rx Spurious emission on ch 0 from 150 kHz to 1 GHz

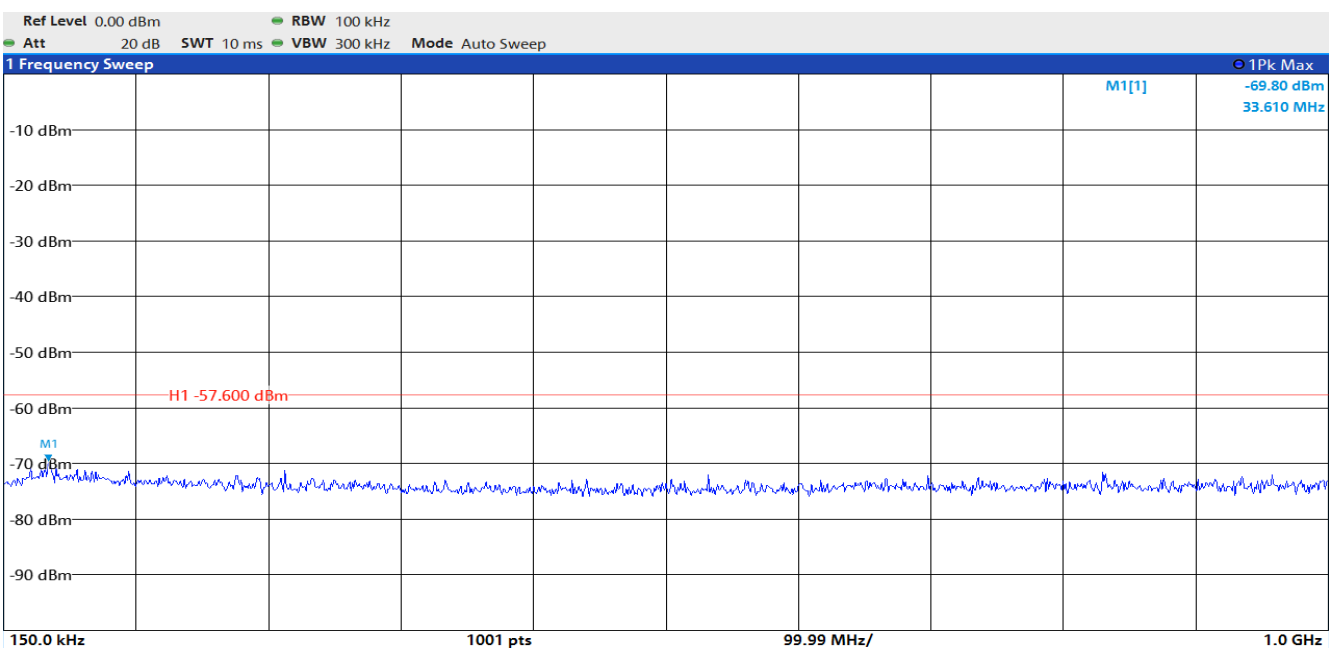


Fig.6.2 Rx Spurious emission on ch 0 from 1 to 12.75 GHz

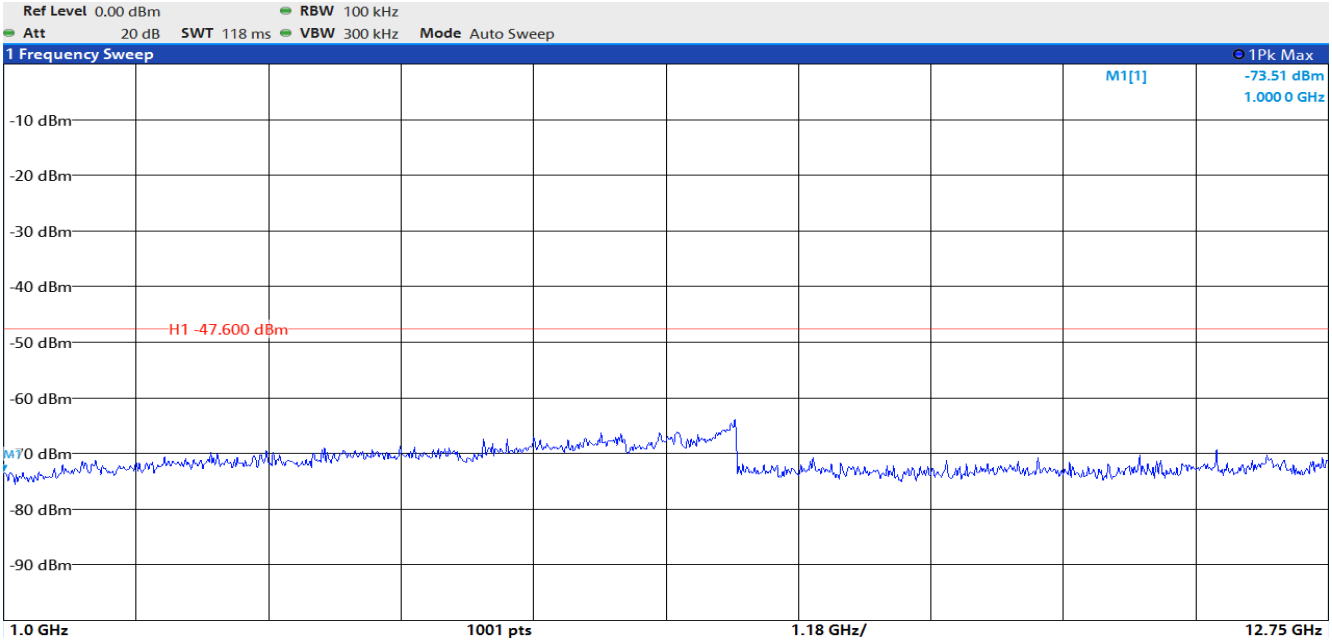


Fig.6.3 Rx Spurious emission on ch 39 from 150 kHz to 1 GHz

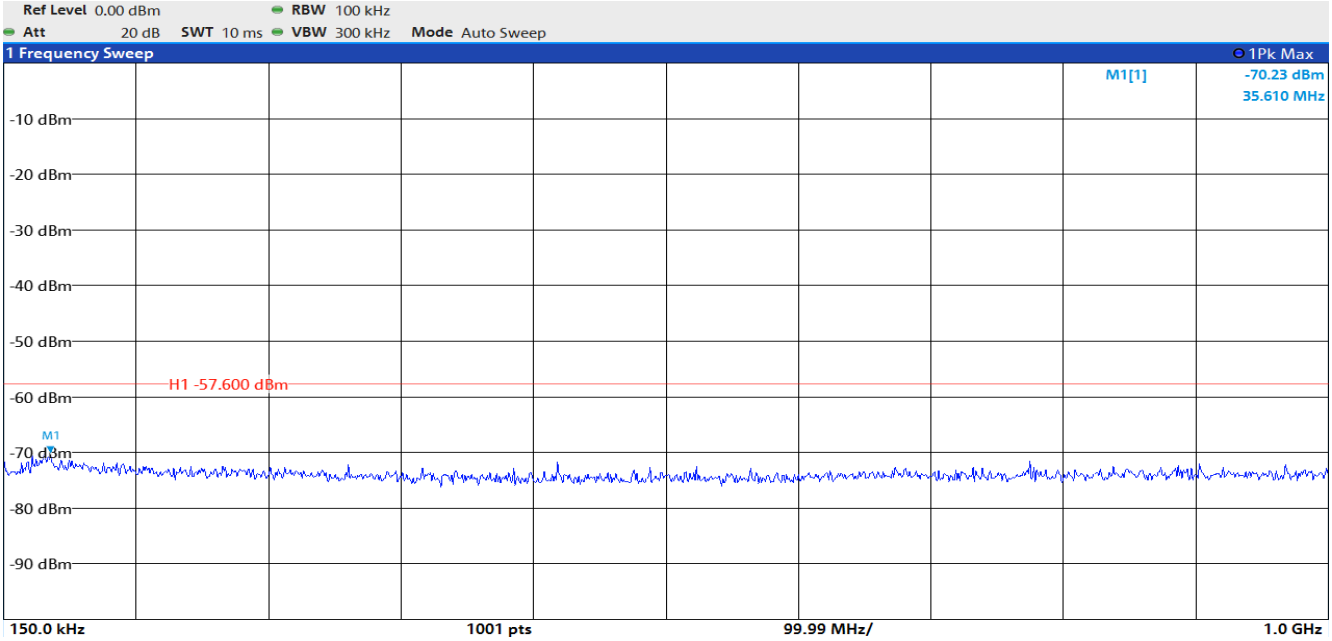
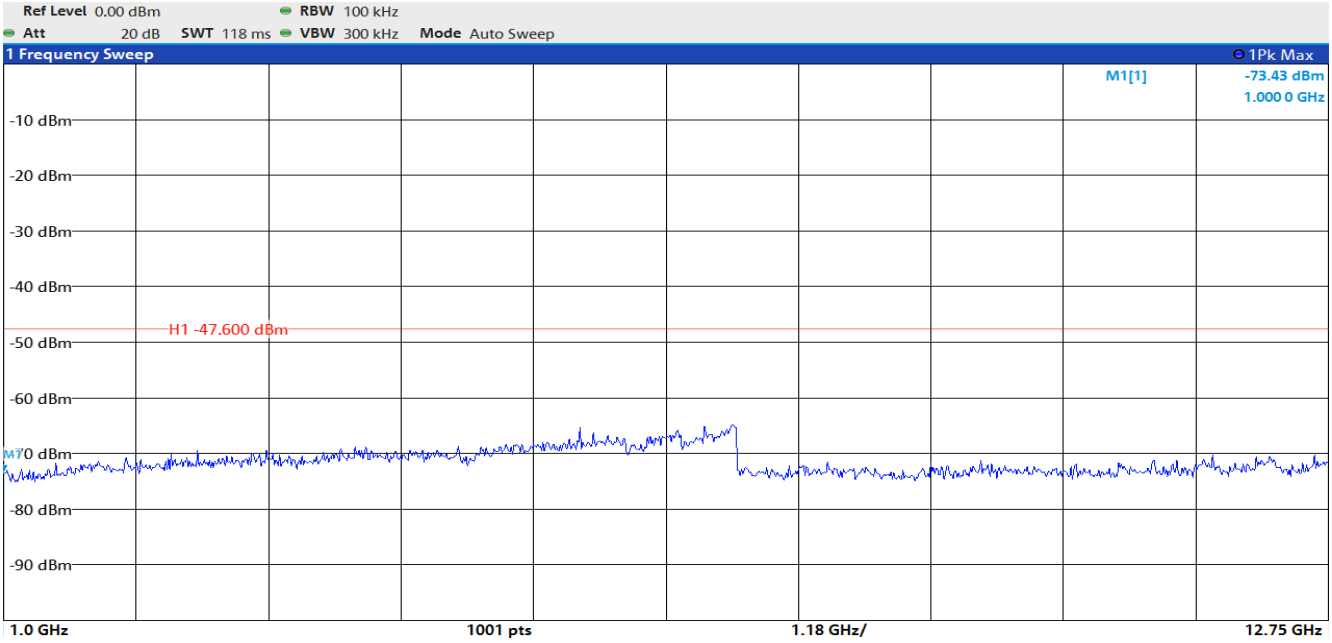


Fig.6.4 Rx Spurious emission on ch 39 from 1 to 12.75 GHz



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